

Tech Newsletter Project

Installation & Setup Guide

1. Overview

This document explains how to install and run the Tech Newsletter project on a new Windows computer.

The guide includes:

- Software requirements
 - Python installation
 - Dependency installation
 - Environment configuration
 - Database setup
 - API configuration
 - Email configuration
 - Running the application
 - Scheduler setup
 - Troubleshooting
-

2. System Requirements

Operating System

- Windows 10 (64-bit)
- Windows 11 (64-bit)

Linux can also be used but this guide targets Windows.

Hardware

Minimum

- Dual Core Processor
- 8 GB RAM
- 5 GB Free Disk Space
- Internet Connection

Recommended

- Intel i5 / Ryzen 5 or better
 - 16 GB RAM
 - SSD Storage
-

3. Software Requirements

Install the following software before running the project.

Python

Version

Python 3.10 or above

Verify

```
python --version
```

or

```
py --version
```

Git

Verify

```
git --version
```

Pip

Verify

```
pip --version
```

4. Project Setup

Extract the project.

Example

```
C:\Projects\tech_newsletter
```

Open Command Prompt or PowerShell.

Move into the project directory.

```
cd C:\Projects\tech_newsletter
```

5. Create Virtual Environment

Create

```
python -m venv venv
```

Activate

Command Prompt

```
venv\Scripts\activate
```

PowerShell

```
venv\Scripts\Activate.ps1
```

Upgrade pip

```
python -m pip install --upgrade pip
```

6. Install Dependencies

Install all required packages.

```
pip install -r requirements.txt
```

Packages include

- feedparser
- requests
- beautifulsoup4
- lxml
- trafilatura
- newspaper3k
- numpy
- scikit-learn
- groq
- google-generativeai
- python-dotenv
- python-dateutil
- schedule
- tzdata

Verify

```
pip list
```

7. Environment Configuration

The project requires a `.env` file.

Create

```
tech_newsletter/.env
```

Example

```
GROQ_API_KEY_1=YOUR_KEY  
GROQ_API_KEY_2=YOUR_KEY  
GROQ_API_KEY_3=YOUR_KEY
```

```
SMTP_HOST=smtp.gmail.com
SMTP_PORT=587
```

```
SMTP_USER=your_email@gmail.com
SMTP_PASSWORD=your_app_password
```

```
SMTP_FROM_EMAIL=your_email@gmail.com
```

Never commit this file to Git.

8. Email Configuration

For Gmail

Enable Two-Step Verification.

Generate an App Password.

Use the App Password instead of your Gmail password.

SMTP Settings

Host

```
smtp.gmail.com
```

Port

```
587
```

Encryption

```
TLS
```

9. API Configuration

The project requires Groq API Keys.

Create one or more keys.

Update the `.env` file.

```
GROQ_API_KEY_1=xxxxxxxxx
```

10. Database

The project uses SQLite.

Database Location

data/newsletter.db

No installation is required.

The application automatically uses this database.

11. Project Directory

tech_newsletter

```
|  
|— card_generator.py  
|— pipeline.py  
|— scheduler.py  
|— manage_employees.py  
|— setup_autostart.py  
|— requirements.txt  
|— README.md  
|— .env  
  
|— config  
|— data  
|— fetchers  
|— filters  
|— generated_cards  
|— llm  
|— logs  
|— newsletter  
|— utils
```

12. Add Newsletter Recipients

Example

```
python manage_employees.py add --id EMP001 --name "John Doe" --email  
john@example.com --dept "IT"
```

Verify

```
python manage_employees.py list
```

13. Test the Application

Dry Run

```
python pipeline.py --dry-run
```

Build Newsletter Only

```
python pipeline.py --no-send
```

Run Complete Pipeline

```
python pipeline.py
```

Debug Mode

```
python pipeline.py --debug
```

14. Scheduler

Run scheduler

```
python scheduler.py
```

Run immediately

```
python scheduler.py --run-now
```

15. Windows Auto Start

Run PowerShell or Command Prompt as Administrator.

Execute

```
python setup_autostart.py
```

Check

```
python setup_autostart.py --status
```

Remove

```
python setup_autostart.py --remove
```

16. Logs

Application logs are stored in

```
logs/
```

Always check logs first when troubleshooting.

17. Generated Files

Newsletter images

```
generated_cards/
```

Database

```
data/newsletter.db
```

18. Verify Installation

Run

```
python pipeline.py --no-send
```

Successful installation should:

- Fetch RSS feeds
 - Extract articles
 - Filter duplicates
 - Score articles
 - Generate newsletter cards
 - Build email
 - Complete without errors
-

19. Common Issues

Module Not Found

```
pip install -r requirements.txt
```

Python Not Recognized

Reinstall Python and enable

Add Python to PATH

SMTP Authentication Failed

- Verify App Password
 - Verify SMTP Host
 - Verify Port
 - Verify Username
-

Invalid API Key

Replace Groq API Key in. env.

Internet Errors

Check:

- Firewall

- Proxy
- Internet connection

Permission Denied

Run terminal as Administrator.

20. Deployment Checklist

- ✓ Python Installed
 - ✓ Virtual Environment Created
 - ✓ Dependencies Installed
 - ✓ .env Configured
 - ✓ SMTP Configured
 - ✓ Groq API Keys Added
 - ✓ Database Available
 - ✓ Recipient Added
 - ✓ Pipeline Tested
 - ✓ Scheduler Tested
 - ✓ Auto-start Configured (Optional)
 - ✓ Logs Verified
-

21. Running the Project

Activate virtual environment

```
venv\Scripts\activate
```

Run newsletter

```
python pipeline.py
```

Run scheduler

```
python scheduler.py
```